

Lecture Unit 3.3

The advection equation

What is advection?

- Advection is the transport of a quantity by a velocity field
- For example the transported quantity could be silt in a river, dust in the atmosphere, salt in the ocean



https://en.wikipedia.org/wiki/Sediment_transport

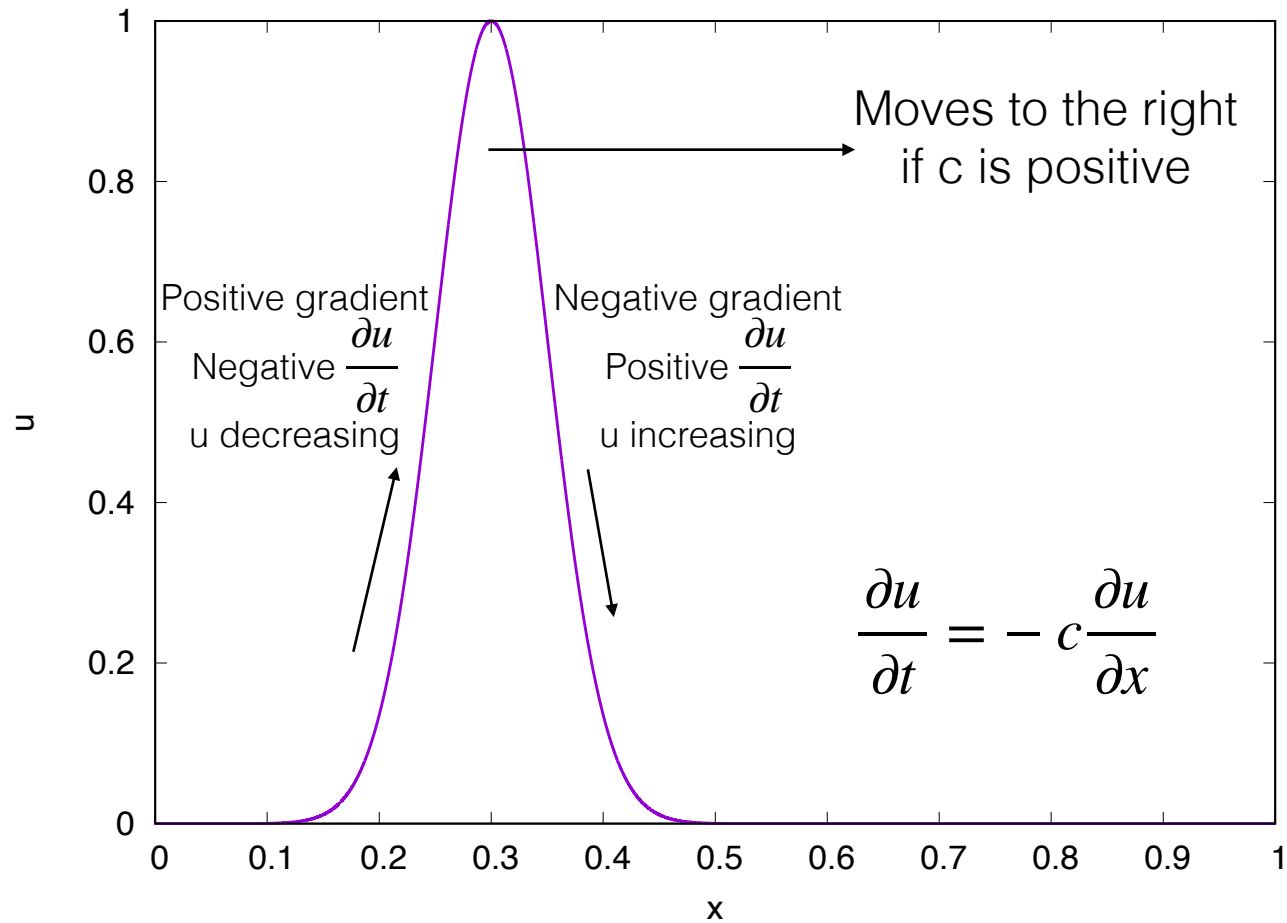
The 1-D advection equation

- The advection of a scalar field $u(x, t)$ by a constant velocity c is described by the advection equation:

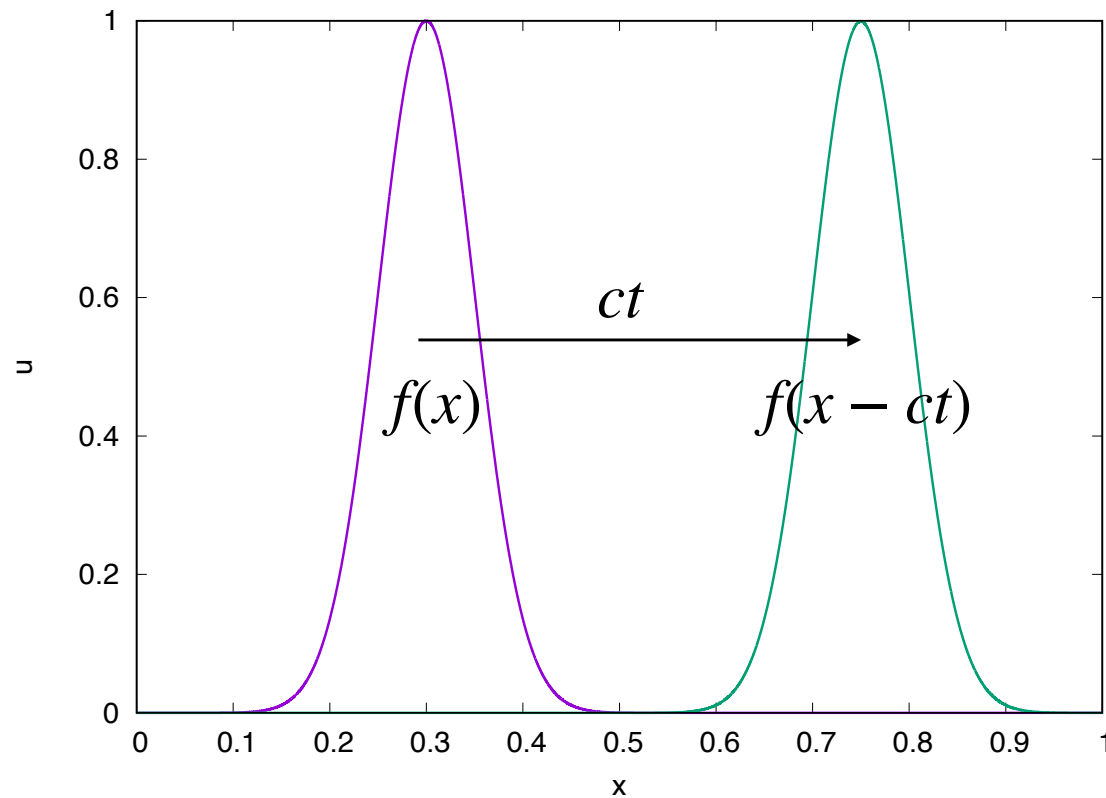
$$\frac{\partial u}{\partial t} = -c \frac{\partial u}{\partial x}$$

- In this equation the right hand side involves a spatial derivative (gradient) of $u(x, t)$

The 1-D advection equation



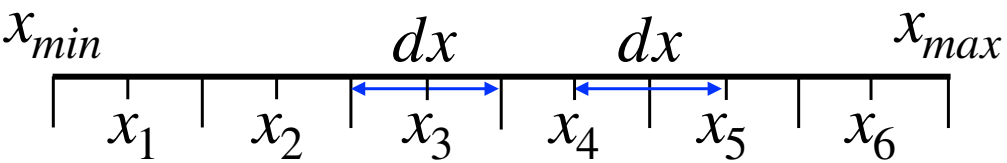
The 1-D advection equation



The initial conditions should translate without changing shape or amplitude

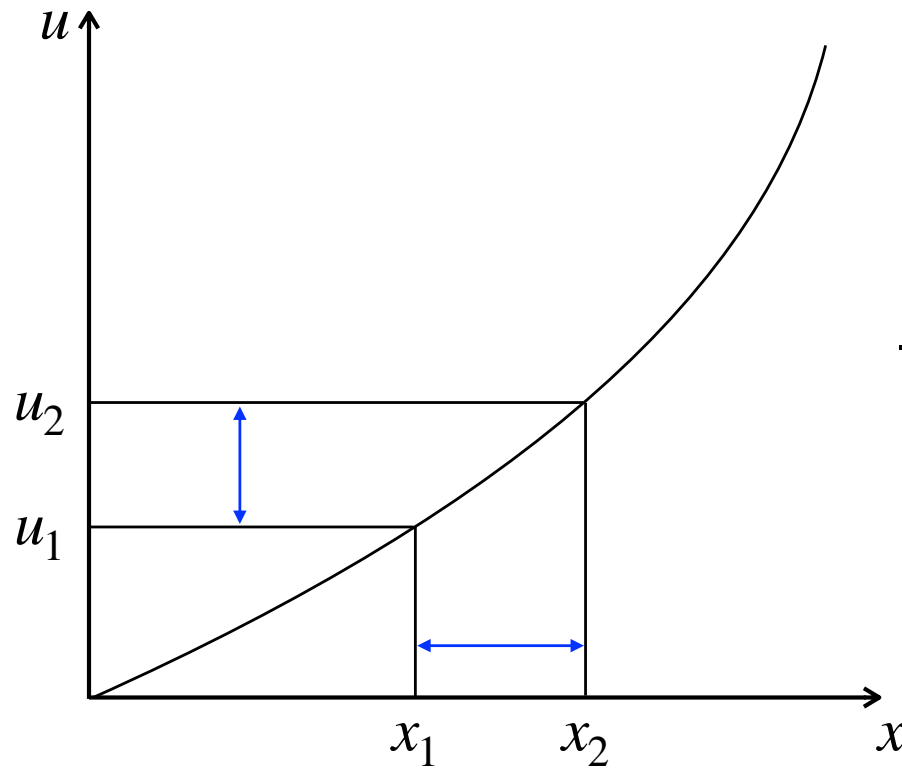
Discretising u

- We can time step the solution like we did for the exponential decay equation but we need a way to calculate $\frac{\partial u}{\partial x}$
- We will represent u using regularly spaced points in space with separation dx
- For example the domain x_{min} to x_{max} divided into 6 intervals:


$$x_i = x_{min} + \left(i - \frac{1}{2}\right) dx \quad u_i = u(x_i)$$

Discretising u

Calculate values for spatial derivatives using differences between grid points:



$$\frac{\partial u}{\partial x} \approx \frac{u_2 - u_1}{x_2 - x_1}$$

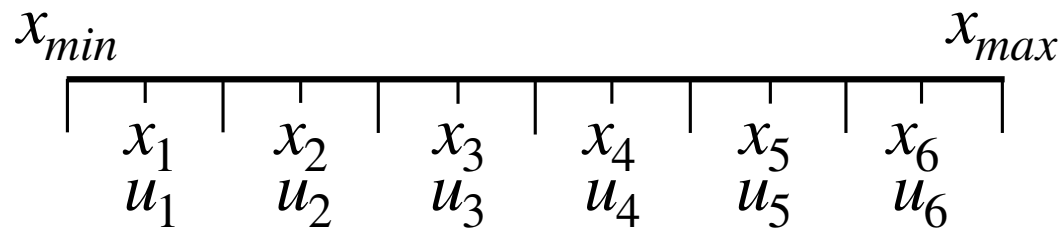
Finite differences

We can use a finite difference expression to calculate $\frac{\partial u}{\partial x}$ at $x = x_i$

Subtract value to the left


$$\left. \frac{\partial u}{\partial x} \right|_{x=x_i} \approx \frac{u_i - u_{i-1}}{x_i - x_{i-1}} = \frac{u_i - u_{i-1}}{dx}$$

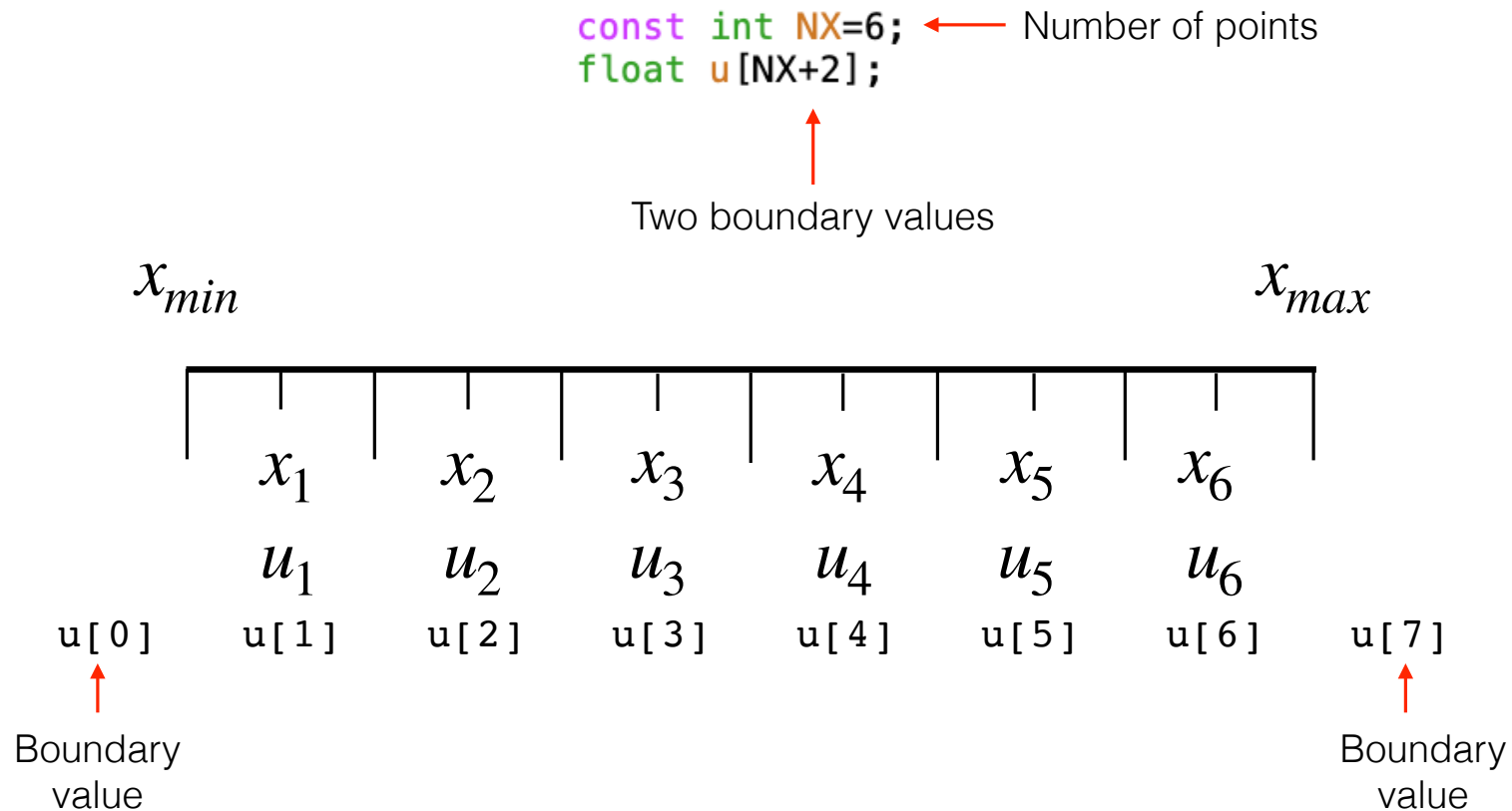
We don't have a value for u_0 at x_0 so what happens at x_1 ?



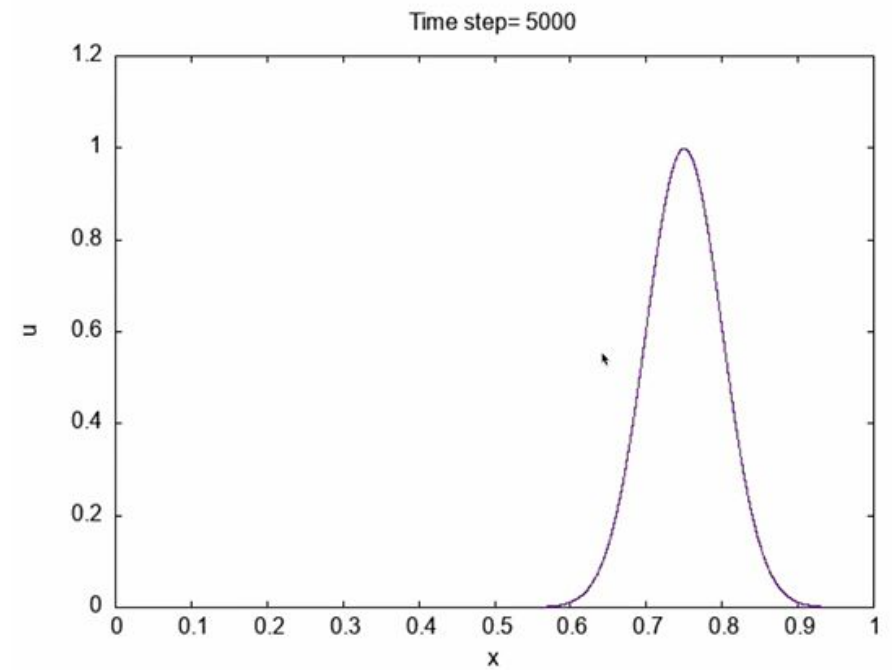
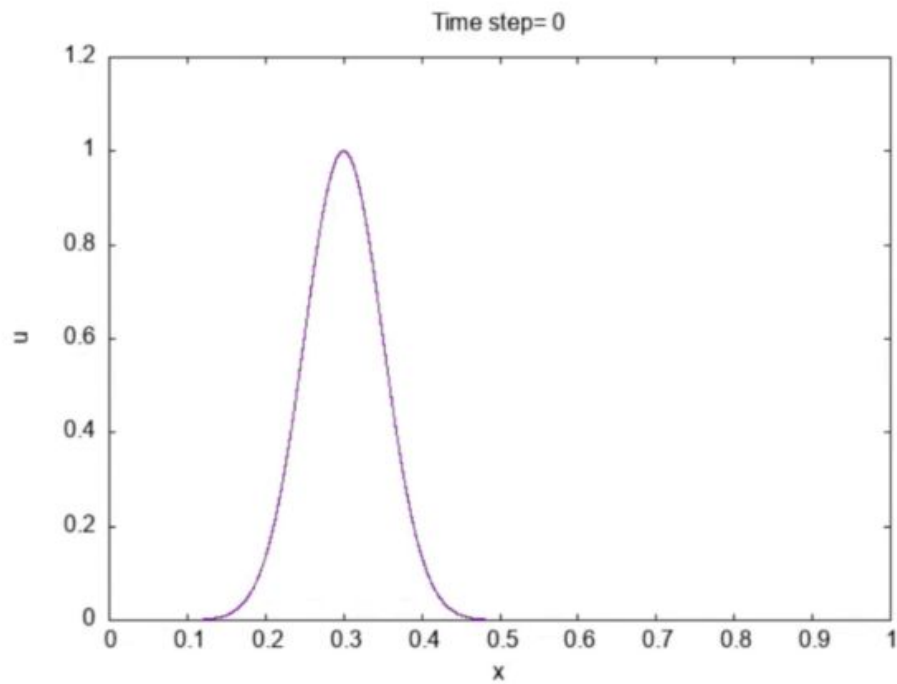
Boundary conditions

- We need to supply extra information so that we have a value for the gradient at x_1
- We can provide a value for the gradient or a value of u to calculate the gradient from
- This extra value is a boundary condition
- Boundary conditions are an important part of specifying the specific problem we are solving
- For example flow over a car could specify velocity and density of upstream air as boundary conditions

Boundary conditions



Boundary conditions



The 2-D advection equation

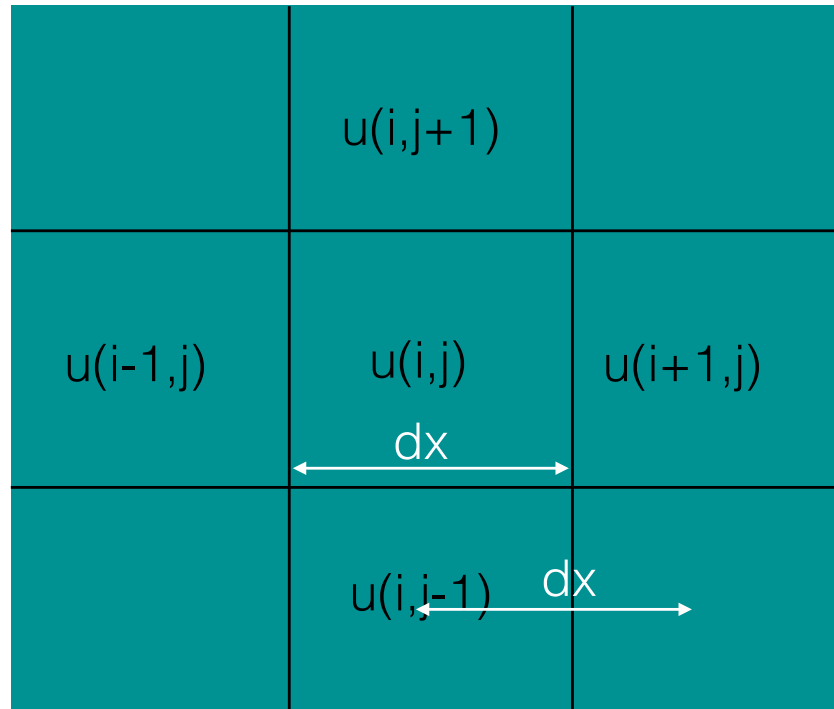
- The advection of a scalar field $u(x, y, t)$ by a constant velocity $\mathbf{c} = (c_x, c_y)$ is described by the equation

$$\frac{\partial u}{\partial t} = - \left(c_x \frac{\partial u}{\partial x} + c_y \frac{\partial u}{\partial y} \right) = - \mathbf{c} \cdot \nabla u$$

- In the 2D case we need to calculate two spatial derivatives but otherwise the solution can be calculated as for the 1D case

The 2-D advection equation

Uniform grid so
cell spacing
is same as
cell width



$$\begin{aligned} \text{dudx}(i,j) &= (u(i,j) - u(i-1,j)) / dx \\ \text{dudy}(i,j) &= (u(i,j) - u(i,j-1)) / dy \end{aligned}$$

Unit Summary

- The advection equation is a partial differential equation containing a time derivative and one or more spatial derivatives
- We can use time stepping to calculate a solution starting from initial conditions like the radioactive decay example
- The spatial derivative can be calculated using a finite difference approximation
- A boundary condition needs to be specified at the edge of the domain